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Data Analytics and Cyber-physical Systems for Smart Manufacturing and Maintenance

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Abstract

This book chapter first discusses a unified Cyber-Physical Systems (CPS) framework for maintenance and service innovation in manufacturing based on a CPS 5C architecture and enabling technologies. Next, a systematic methodology and corresponding advanced data analytics are proposed to build Digital Twins (DT) for intelligent maintenance and service. Then, two real-world cases are provided to demonstrate the proposed framework and methodology. Finally, the existing challenges and future works are discussed.

Key Words

Cyber-Physical Systems, Smart Manufacturing, Digital Twins, 5C Architecture, Intelligent Maintenance, Prognostics and Health Management, Data Analytics, Enabling Technologies

1. Introduction

Maintenance and service innovation is an essential topic in manufacturing that can help practitioners improve the quality of products and services, optimize asset management, reduce operation costs, and, therefore, maintain their competitiveness in the market. In recent decades, the fast growth of sensor technology, internet connectivity, and advanced data analytics have radically transformed maintenance practices in manufacturing from fail-and-fix to predict-and-prevent, with the target of reducing unplanned downtime and maintenance events. Such a transformation was incentivized by the rapid growth of condition-based maintenance (CBM) and prognostics and health management (PHM) in recent years, which aim to avoid costly failures and unexpected downtime of machinery by uncovering invisible degradation patterns in the operational data, identifying anomalous product manufacturing processes, and predicting remaining useful life of the machine. In CBM and PHM, machine connectivity, data collection, and advanced data analytics play an essential role in making the machines aware of potential pending failures and enabling intelligence maintenance and service innovations. Currently, the CBM and PHM technologies have been widely adopted in many manufacturing sectors, such as semiconductor manufacturing [1], material

handling [2,3], injection molding, additive manufacturing, micro-manufacturing, etc., and numerous successful use cases are reported in the literature [1–11].

As the CBM and PHM technologies continue to grow, maintenance and service innovation is playing a more and more important role in manufacturing. According to the National Institute of Standards and Technology's (NIST) economic analysis brief in 2016 [12], novel intelligent manufacturing technology can save US manufacturers \$57.4 billion annually. Among all the capabilities that smart manufacturing enables, enhanced sensing and monitoring have the largest projected economic impact which is up to \$10,910 million annually. Seamless transmission of digital information and advances in data analysis have a projected annual impact of \$ 10,310 million and \$ 10,134 million, respectively. These numbers do not account for the economic impact of new products' commercialization, research and development cost savings, and other societal impacts in terms of long-term technological growth. Moreover, compared with the declining manufacturing jobs in the US (from 17 million in 2000 to the lowest point of 11.4 million in 2010), the services jobs in the sector of professional, scientific, and technical services are growing from 7.5 million in 2009 to nearly 10 million in 2020 with a cumulative growth of over 30%. Likewise, the purchases of repair and maintenance services for machinery have increased by over 30% since 2009. In 2020, these purchases contributed \$ 2.3 trillion to the US economy, which has outpaced the \$ 2 trillion contributed by manufacturing [13].

CPS and advanced data analytics are the key elements that will drive future maintenance and service innovations in manufacturing. CPS are engineered systems that exploit the integration of computation and physics machines to transform the concepts of manufacturing. The implementation of CPS in manufacturing aims to enable smart and digital manufacturing, advanced industrial robotics, and new infrastructure for AI and cybersecurity. It converges advanced sensors and actuators, fast and secure data transmission, edge and cloud computing, advanced data analytics and AI, advanced control, etc. Among these elements, advanced data analytics and AI are the key enablers of providing smart services, which process machine data into actionable knowledge and decisions and then allow computers or automated systems to perform tasks that have historically required human cognition and decision-making capabilities. In practice, these advanced data analytics and AI algorithms are combined with other data processing steps, such as data granulation and feature extraction, among others, to build a digital replica of the manufacturing systems, processes, and equipment that are also known as DT. The DT takes the real-time operation data as input and generates meaningful information and decisions for control and management actions. Typical applications of the DT in manufacturing include virtual metrology, fault detection & classification, machine Remaining Useful Life (RUL) prediction, planning and scheduling, yield prediction, knowledge retention and expansion, risk quantification, etc.

The widespread adaption of CPS technology and advanced data analytics will open new opportunities for maintenance and service innovations which include, but are not limited to, 1) maintenance of large fleets of geographically distributed manufacturing assets (e.g., machine tools, injection molding machines, etc.); 2) using advanced analytics to describe complex physical dynamics in manufacturing systems and processes, and discovering the invisible relationship between process data and product quality; 3) improving the manufacturing resiliency when the factors such as demand, capacity, inventory, and manpower are highly interconnected; 4) facilitating the transformation from optimization within the four walls to the optimization across the entire supply chain, including factors such as market fluctuations and customer experience.

This book chapter first discusses a unified CPS framework for maintenance and service innovation in manufacturing. Systematic methodologies and advanced data analytics for building DTs in the proposed CPS framework are then introduced. Two real-world use cases are provided to demonstrate how this framework is applied. Finally, we highlight the existing challenges and future works.

2. CPS Framework for Maintenance and Service Innovation

2.1. 5C Architecture for CPS Framework

CPS is smart and transformative technology for implementing the thorough process of data-related operations such as acquisition, aggregation, analysis, prediction, deployment, and decision making. The requirement of logistics, the production process interaction, and the supply chain are fully considered. Therefore, CPS is the key to the successful deployment of maintenance and service innovation. However, since CPS is still at the early stage of development, a clear definition of the structure and methodology of a CPS is critical to clarify for industry practitioners. A unified framework has been proposed in [14] to explicit the details. A 5-level CPS architecture consisting of a connection, conversion, cyber, cognition, and configuration layer called 5C is proposed, as shown in Fig. 1. It provides a practical method for developing and deploying CPS in industrial applications. A detailed description of each layer is provided as follows:

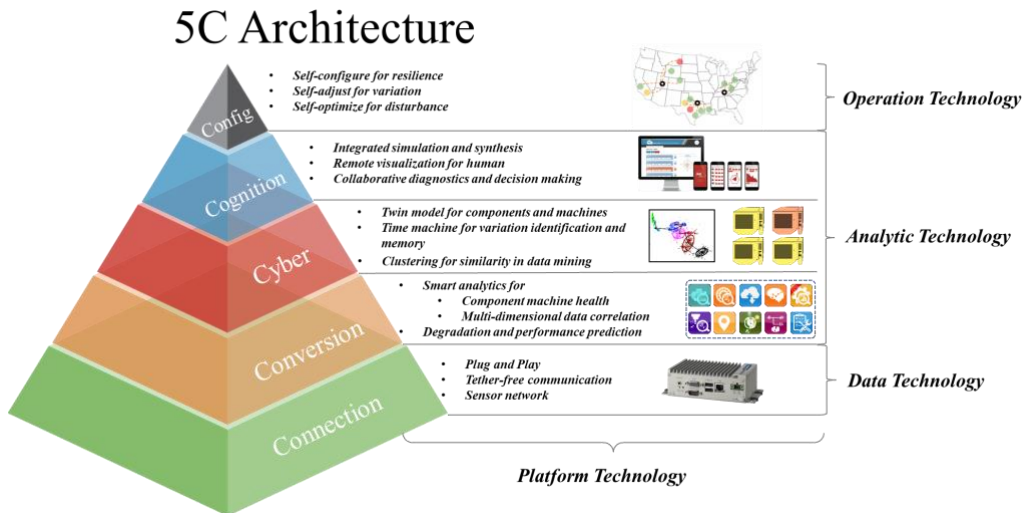


Fig. 1. CPS 5C architecture and the enabling technologies.
Credit to [14].

Connection layer: The connection layer is the foundation of the whole 5C architecture. It incorporates smart sensing technology to ensure the data quality and completeness of the information sources, such as sensors, controllers, or enterprise manufacturing systems. The interconnected data environment and data acquisition channels are established to support the upper layers of CPS. Besides, another key aspect of smart sensing in this layer can select desired data acquisition due to the customized target and needs of information analysis. The collected data will be seamlessly transferred to the server through interconnected channels with unified communication protocols and standards, such as MT-Connect. Selecting the appropriate smart sensing technologies in this step will significantly impact the overall performance of the CPS throughout the next levels.

Conversion layer: The conversion layer emphasizes converting raw data into useful information using smart analytics for individual components or machines. Such analytic technologies include feature extraction, filtering, and prioritization of the data with the integration of the domain knowledge of the expertise. Multi-dimensional data correlation analysis, data-driven health assessment, degradation, and performance prediction for an individual component or machine are

common tasks in the conversion layer. The implemented analytics will guarantee the interpretability of extracted information, which prepare for the advanced analytics in the next layer of 5C architecture. The obtained useful information represents complex physical dynamics in assets that will be pushed into cyberspace.

Cyber layer: The cyber layer acts as a central information hub in the 5C architecture. Massive information obtained from the conversion layer is pushed to the infrastructure in this layer and used for higher-level analytics such as building twin model, time machine, and clustering for similarity. Rather than dealing with raw data as in the conversion layer, the cyber layer takes information from multiple sources for synchronization, modeling, analysis, and prediction at a higher system level. Advanced analytics can be used for extracting insight into the condition of individual machines among the fleet, providing machines with the self-comparison ability. Similarities between machine performance and previous assets can be compared to estimate the future behavior of machines. Once the cyberinfrastructure of the machines is implemented, machines can be enrolled into the network to share information. It can be seen as a time machine and provide three functions as follows:

1) Snapshot information collection:

When the machine information is uploaded to cyberspace, only the most valuable information, as opposed to the whole time series, will be managed and stored in an efficient way to reduce the space loading. When an important change has occurred to the machine, the snapshot will store the current status. The snapshot itself will be accumulated during the machine's life cycle, providing the availability of history tracking and peer-to-peer comparison. When these events occur, the active status of the snapshot is changed to historical and used for functions such as similarity identification, and synthesis utilization, respectively.

2) Similarity identification and machine clustering:

As the information among machines can be obtained in the cyberinfrastructure, identifying a certain failure pattern becomes possible. It refers to the historical record to compare the patterns between current and former assets via similarity computation. When the most similar pattern is matched, the future trend of the current asset can be accurately predicted.

3) Synthesis optimized utilization:

With the aid of prediction techniques, a just-in-time maintenance strategy can be developed for industry practitioners. Maintenance operation and asset utilization efficiency can be optimized based on the predicted result.

Cognition layer: The implementation of the cognition layer creates a comprehensive knowledge of the monitored system. Proper visualization of the generated knowledge supports the expertise and users to make the appropriate decision. Since the comparative information and the condition of individual machines can be accessed, the decision on the task priority of different machines can be made.

Configuration layer: The configuration level in the CPS architecture is the response from cyberspace to physical space and acts as supervisory control to adaptively adjust the configuration of machines. According to the maintenance task and the analysis parameters in the cognition layer, it performs the corrective and preventive operations, which have been decided from the previous cognition layer to the monitored system.

2.2. Enabling Technologies for CPS Framework

Some key enabling technologies are summarized in order to help practitioners achieve success in the mentioned Connection, Conversion, Cyber, Cognition, and Configuration. Such enabling

technologies are outlined in Data Technology, Analytical Technology, Operation Technology, and Platform Technology, which are elaborated as follows:

Data Technology: The unique data is generated in each step during the manufacturing process. These steps in the individual process can generate a great number of valuable data at either the component level, the machine level, or the shop-floor level. It can be categorized as structured and unstructured data. The 3B data problems, namely broken data, bad data, and background of the data, have a massive impact on implementing the CPS 5C architecture in industrial applications. Apart from achieving the unified collection of heterogeneous data, achieving the automatic extraction of valuable information and enhancing the standardization of data acquisition is also crucial. Besides, data management and synchronization should also be considered. Several core aspects of Data Technology include 1) sensor technology and data acquisition, 2) data indexing, transmission, and storage, 3) data quality evaluation, 4) big data representation and visualization, 5) data security and privacy protection, etc.

Analytics Technology: Sensors and data acquisition systems in the data technology can provide the substantial data that need to be analysed to extract valuable information. Analytics technologies act as the interface between the physical and cyber worlds. It corresponds to the conversion layer in the CPS framework, including pipeline analysis at the edge and cloud computing. Even though the edge end is closer to the industrial site and has a good ability to analyse real-time data, it is difficult to take centralized model training and prediction due to the limited storage space and data processing capabilities of edge computers. Therefore, based on the inherent nature of edge computing, it is necessary to focus on the innovation in flow-based reasoning technologies and use the real-time data stream to recursively refine the data relevance to meet model self-optimization. New modeling and analysis tools need to be explored like time machine-based state modeling, semi-supervised learning, peer-to-peer learning, adaptive learning, and transfer learning. These emerging techniques consist of 1) signal processing and statistical analysis, 2) detection, classification, prediction, clustering, relationship-mining, 3) causal analysis or inference, 4) modeling complex and dynamic systems, etc.

Operation Technology: Operation technologies refer to making decisions and taking actions based on information. They are used in the cognition layer of the CPS 5C architecture. In conjunction with three other technologies, it is used to optimize production during the manufacturing process. It embraces three aspects in the development of operational management methods: 1) Effective transformation of the acquired knowledge from the prediction model into operational maintenance, 2) management decision-making at the enterprise level and 3) the implementation of the transformation from experience-driven production to data-driven production.

Platform Technology: Platform technologies act as the carrier of all other enabling technologies and support the corresponding function of the connection layer, conversion layer, and cyber layer in the CPS architecture. Likewise, for the edge computing platform, it is necessary to enhance the ability of data acquisition and computing via the innovation in hardware technology as well as developing the collaborative capability of edge platforms to realize self-organization and self-configuration in the production system. The interconnection property of platforms calls for cybersecurity. Hardware architecture and software mechanisms must ensure the security of system access, especially after the feedback control of the edge device is implemented. For cloud platforms, it is crucial to develop several key elements: 1) life cycle management model, 2) assisting uncertainty management model, and 3) continuous self-learning model. In general, platform technologies can be divided into three levels of services: Infrastructure as a Service, Cloud as a Service, and Solutions as a Service.

2.3. CPS Architecture for Maintenance & Service Innovation

The implementation of CPS 5C architecture could innovate maintenance and service in industrial applications [14,15]. Fig. 2. is an illustration of such implementation of CPS 5C architecture for

machine tools intelligent maintenance. At the connection level, data is collected via add-on sensors and controllers, thereafter, transferred to the local data server. Then, the sensor data from the local server is converted to valuable information and uploaded to cyberspace. At the cyber level, the cloud server executes the adaptive method to cluster the machine status by comparing the similarity between current and former machines. This health status could be further utilized in the cognition and configuration level. For example, the important health information of the machine can be transferred to the end-user for decision-making via a Web-based user interface or mobile application. Therefore, the significant benefit at three different stages is summarized. At the component stage, once the valuable information has been extracted from the sensory data in the local data server, a DT of each component is used to track machine records as well as predict future steps to provide the capability of self-awareness and self-prediction. At the machine stage, additional supplementary information, such as controller parameters and the adjustment record, will be integrated to form the cyber-twin of each machine, providing the capability of self-comparison. At the final production stage, the accumulated knowledge and information provide self-configurability and self-maintainability to the factory and help users optimize the production and management plan in a real-time way via the internet or mobile application.

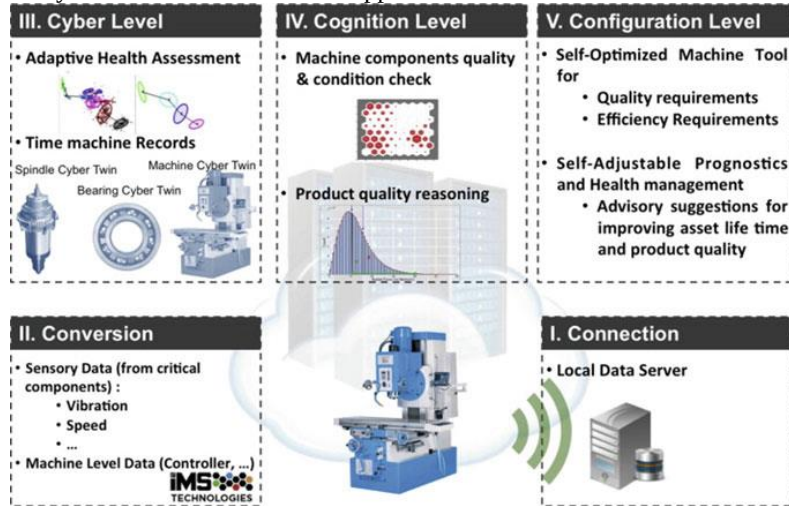


Fig. 2. An example of CPS 5C architecture implementation for machine tool intelligent maintenance. Credit to [14].

3. Development of Digital Twins: Methodology and Analytics

3.1. Digital Twin for Maintenance & Service Innovation:

DTs form the basis of the CPS system and are the key elements in the conversion layer and cyber layer of 5C architecture. In the manufacturing sector, DTs are described as purpose-driven digital replicas of components, assets, and manufacturing systems and processes. Michael Grieves first introduces the concept of DT in his discussion about Product Lifecycle Management (PLM) in 2002 [16]. Since then, different industries have adjusted the definition of DT in their specific perspectives. Table 1 lists the definitions of DTs given by the leading enterprises in the manufacturing sector.

In practice, the DTs are mainly used to process the massive machine data into meaningful information and timely decision to achieve improved business outcomes. It incorporates the bottom-up knowledge gained from the development and implementation of DT solutions (e.g., Predictive Maintenance (PdM), VM, Model-Based Process Control (MBPC), etc.) The design requirements

for an effective DT include reusability, extensibility, interoperability, interchangeability, and autonomy, as discussed in [17]. The applications of DTs for maintenance and service innovation in manufacturing are mainly listed in Table 2, which include production scheduling and planning, maintenance and monitoring, cyber-security, supply chain optimization, design improvement, etc.

As illustrated in Fig. 3, the methodology for developing and deploying the DT solution can be summarized in two phases. Phase 1 focuses on offline DT model development from scratch. Phase 2 emphasizes deploying, using, and updating the developed DT model in the online scenario. The objective is to empower the physical machine with intelligent maintenance capability based on the CPS 5C framework.

Table 1. Definitions of DTs from the leading enterprises in the manufacturing sector

Organization	Definitions of Digital Twin
IBM	A DT is a virtual representation of an object or system that spans its lifecycle, is updated from real-time data, and uses simulation, machine learning, and reasoning to help decision-making [18].
TWI global	A DT is a digital representation of a physical object, process, or service. A DT can be a digital replica of an object in the physical world, such as a jet engine or wind farm, or even larger items such as buildings or even whole cities [19].
GE digital	DT is most commonly defined as a software representation of a physical asset, system, or process designed to detect, prevent, predict, and optimize through real-time analytics to deliver business value [20].
Siemens	A DT is a virtual representation of a physical product or process, used to understand and predict the physical counterpart's performance characteristics. Digital twins are used throughout the product lifecycle to simulate, predict, and optimize the product and production system before investing in physical prototypes and assets [21].
Schneider Electric	A DT is not a result, product, outcome, or technology in and of itself. Instead, it is a dynamic data-supported framework that functions as a business enabler leading to results, products, outcomes, or new technologies. It is a means to an end, and the end is solving a real-world problem with real-world data [22].

Table 2. Applications of DTs for maintenance and service innovation

Type of DT	Description	Example
Production Planning and Scheduling	A different section of the production process can be optimized using a digital twin	Using DT for the production line of a 'Solo' electric car of Electra Meccanica [23]
Maintenance and Monitoring	DT frameworks can be exploited to improve different services e.g., maintenance and monitoring	Airbus DT model, which provides predictive maintenance services [24]
Cyber-security	The security aspect of any system from physical to cyber level can be modeled and optimized using a digital twin	Haruspex cyber security DT [25]
Supply Chain Optimization	A virtual DT can model assets, logistics, warehouses, and inventory status	Unilever PLC DT for supply chain management [26]
Design Improvement	Design principles and parameters can be optimized using the interaction of digital and physical spaces	GE plant optimizer [27]
Safety Planning	The gaps in the safety perspectives can be bridged using a DT framework. These safety gaps may be addressed in physical space, separately	Collaboration between Energy Department's Oak Ridge National Laboratory and National Renewable Energy Laboratory for developing a DT to optimize the driver's safety principles [28]
Collaboration	Different types of collaboration, e.g., human-to-machine, human-to-human,	Philip heart DT [29]

	and machine-to-machine	
Knowledge Retention and Expansion	Defining new types of subject matter experts (SMEs)	Worley DT for BP Clair Ridge [30]

3.2. Methodology for Digital Twin Development and Deployment (D&D)

Fig. 3 outlines a methodology for developing and deploying the DT solution. The proposed methodology comprises two phases. Phase I focuses on offline DT model development, including DT Modeling, DT Adaptation, and DT Validation and Verification (V&V). Phase II addresses the challenges of updating DT online after deployment, such as identifying the critical samples that need human labels, monitoring the performance of DTs after deployment, retaining the model performance by learning the up-to-date machine data, etc.

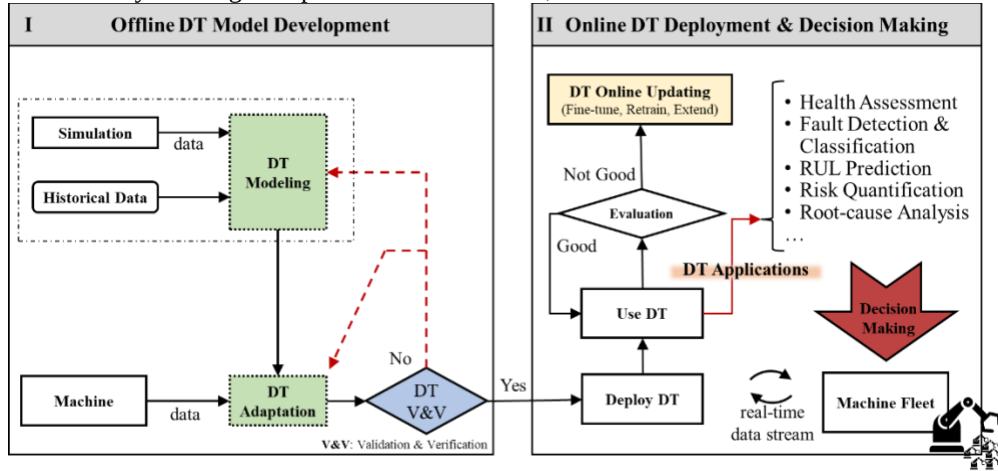


Fig. 3. The methodology of developing and deploying DT model for intelligent maintenance and services. Credit to this book chapter.

Phase I-DT Development: This phase aims to train a DT model that could be used to monitor, analyse, predict, and optimize manufacturing performance [17]. The DT developments phase has three major steps, which are as follows,

A) DT Modeling: The goal of this step is to develop a useable DT (i.e., a series of data processing and analysis procedures) as a virtual representation of the physical components, assets, systems, and processes. The expected DT model must be generally applied to a cluster of target machines or processes under specified conditions. The integration of physical knowledge, engineering experience, and data insights must be highlighted to ensure the generality of the DT model. Also, the training data used for DT modeling should be generated from computer simulations or the design of experiments that hold a good coverage of parameter space to be explored. In other words, the training data must have enough variations (e.g., different degradations stages of the machines, multiple common operation conditions, etc.) to build a useable DT model. To give an example, Fixed Cycle Feature Test (FCFT), which commands the machine to operate under a set of pre-designed speed levels and loading conditions, is often adopted to generate representative data for machine health monitoring and prediction. In injection molding, computer experiments are often used to generate datasets for process control optimizations.

B) DT Adaptation: The goal of model adaptation is to adapt the general DT model based on the historical data that is collected from the individual machine or specific manufacturing process. Compared with the DT modeling step that aims to improve the model's generality, the DT adaptation steps enhance the practicality and individuality of the DT model (i.e., the performance/accuracy of

the DT model when it is applied to individual machines or manufacturing processes). The DT adaptation is a critical step when the machine-to-machine difference is significant or when the target machine is frequently operated under certain operating conditions.

C) DT Validation and Verification: The DT validation and verification (V&V) step aims to evaluate the credibility of the DT model for deployment. For model validation, the model performance, including accuracy, agility, robustness, etc., must be tested against the historical data and experimental data from the physical machines or processes. The behavior of the DT model under extreme conditions must be considered. The degeneracy of the model performance must be extensively tested by removing portions of the model or by selecting values of inputs and internal parameters. The model verification ensures the given computation program correctly describes a mathematical model over a state range of inputs. The development procedures from software engineering, such as top-down design, program modularity, etc., should be followed. In this step, if the DT model passes all the DT V&V procedures, it will be routed to the next step for online deployment. Otherwise, the DT modeling and adaptation steps will be repeated until the model is ready.

Phase II-DT Deployment and Online Updating: This step aims to deploy, monitor, and update the developed DT model in the online scenario to ensure sustainable performance. Unlike offline development, this phase emphasizes using real-time (or nearly real-time) data from its operating environment to retain and refresh the model performance.

A) DT Applications: In the online environment, the machine data is continuously pushed to the DT model for analysis. The role of DT is to process the (near) real-time machine data into meaningful information and decisions efficiently. Typical applications of the DT in manufacturing settings include health assessment, RUL prediction, risk quantification, root-cause analysis, yield prediction, virtual metrology, etc.

B) DT Monitoring: The goal of DT monitoring is to continuously monitor the Key Performance Indicators (KPI) of the DT model, such as prediction accuracy, model robustness, etc. The KPI of the DT model describes the performance of the DT model over a certain period or a specific set of data samples. DT monitoring is important because the data shift/drift, often induced by invisible factors like human errors, maintenance activities, ambient disturbances, etc., could cause the model performance to degrade rapidly. When the KPI of the model starts to decrease, the important samples that can bring the biggest improvement to the model need to be identified and augmented to the training data.

C) DT Online Updating: DT online updating aims to retain the model performance by dynamic tracking the data distribution changes (i.e., data shift/drift) in an online environment. The online model updating scheme comprises several important steps: 1) obtain a confidence indicator or reliance indicator for the model prediction; 2) identify the critical samples to get human inputs; 3) update the model parameters based on the new data through model re-training or online learning mechanisms; 4) deploy the new model after validating the performance.

The widespread adaptation of DTs has the potential to revolutionize maintenance and service in manufacturing by enabling efficient fleet-sourced data analysis and decision-making. In a networked environment, the individual DT servers an agent to aggregate the information and decisions from similar peer machines or upstream/downstream manufacturing processes for improved outcomes. For example, collecting lifecycle data for PHM modeling is extremely expensive and time-consuming because the time between failures can be very long. As such, the individual machine is likely to suffer limits in the data, leaving it unable to build and deploy PHM models with acceptable performance and efficiency. In this case, the peer-to-peer analysis can allow timely information/knowledge-sharing across fleet networks and by predicting the machine failures with improved accuracy. Likewise, integrating the DTs of multiple manufacturing stages across the

production line enables improved yield prediction and root-cause backtracking, which is critical in improving product quality.

3.3. Data Analytics for Digital Twin Development and Deployment

DT relies on advanced data analytics to describe the complex physical dynamics of manufacturing machines, systems, and processes. Advanced data analytics could generate insights and suggest decisions for different control or management applications. In the following discussion, we will review the frequently used data analytics for DT modeling, DT adaption, DT validation and verification, and DT online monitoring and updating:

A) DT Modeling: The advanced analytics related to DT modeling include the Design of Experiments (DoE) methods, feature extraction, and signal processing algorithms, data quality evaluation, and physics-informed machine learning techniques.

- *Design of Experiments Methods*, such as space-filling design, Taguchi methods, FCFT, etc., are often used to collect representative data samples that cover a stated range of input variables or a series of operating conditions. In case when experiments are costly or unavailable, computer experiments based on simulation programs can be planned. The data from computer experiments can be utilized to develop an efficient surrogate model of the simulator.
- *Feature Engineering* algorithms process the machine data into important features (i.e., compact representations of the raw data). In many applications, the desirable feature representation holds maximal relevance to the modeling target and minimal redundancy among features. Descriptive statistics (e.g., mean, variance, root-mean-square, Kurtosis, etc.) and signal processing techniques (e.g., wavelet analysis, envelope analysis, etc.) are often used to extract useful information from the raw data. Dimension reduction techniques (e.g., principal component analysis, t-distributed stochastic neighbor embedding, etc.) and feature selection algorithms (e.g., recursive feature elimination, lasso, hill-climbing search, etc.) are often used to minimize the redundancy of the extracted features.
- *Feature/Data Quality Evaluation* is important to confirm the suitability of the dataset for modeling (i.e., if the current data carries important information for modeling). The data quality evaluation can be based on either the raw data or the extracted features. Based on different modeling purposes, the detectability, clustering tendency, predictability, and traceability of the data should be evaluated. For time-series data, the signal characteristics (e.g., periodicity, randomness, spikiness, etc.) can be also evaluated.
- *Physics-Informed Modeling Techniques* refers to the data analytics that is used to describe the behavior of physical machines or processes. Depending on the DT applications, the commonly used data-driven analytics for modeling are tabulated in Table 3. The use of first-principle models is also very common in practice. However, these models are often developed for specific applications based on experiments or simulations.

Table 3. Commonly used data-driven analytics for modeling

DT Application	Algorithms
Health Assessment	Logistic Regression [31]
	Self-Organizing Map [32] [33]
	Deep Neural Networks [6]
Fault Detection & Classification	Support Vector Machine [34]
	Naive Bayes
	Deep Neural Networks [35]
	Fuzzy Logic [36]
Remaining Useful Life Prediction	Deep Neural Networks[6]

	Similarity-Based Prediction Methods [37]
	State Space Model[38]
	Random Coefficient Model
Risk Quantification	Weibull Analysis [39]
	Copula Models [40]
	Gaussian Process Models
	Bayesian Theory [41]
Root Cause Analysis	Bayesian Belief Networks [41]

B) *DT Adaptation* is also called DT individualization, which aims to adopt a generalized DT model to a specific use case. Transfer learning algorithms can be utilized when abundant historical data from the target machine is available. When limited historical data from the target machine is observed, the Bayesian learning algorithm and non-parametric learning algorithms can be implemented to adapt the model based on a small set of data samples.

- *Transfer Learning* algorithms can significantly improve the reusability of the existing DT solutions or datasets by promoting knowledge transfer across specific modeling tasks. It, therefore, maximizes the value of data and DT solutions that have been historically developed. For example, transfer learning is used to individualize a generalized DT solution to a specific machine or process for improved performance. It is also used to overcome machine-to-machine differences by transferring an existing DT solution from one machine to another one. The commonly used transfer learning algorithm includes the deep neural networks regularized by the Maximum mean discrepancy and the kernel-based methods (e.g., transfer component analysis, joint distribution analysis, domain-invariant component analysis, etc.).
- *Non-Parametric Methods* (e.g., k-nearest neighbor, Gaussian process model, kernel machines, support vector machines, etc.) make predictions based on the adjacency between the incoming data and the historical observations. The non-parametric methods do not have prior assumptions of functional forms for model training. Therefore, these algorithms are flexible to data distribution shifts and drifts by simply adding/removing the samples in the training data. For example, the decision boundary of a support vector classifier will be changed once the support vector set is updated. The Gaussian process regression can adapt to the data drifts by simply replacing the training data with new data samples without updating the hyper-parameters.

C) *DT Validation and Verification* analytics seek to quantify how well the DT model represents the physics from the source and target domain. Methodologies include model performance evaluation, sensitivity analysis, model behavior under extreme conditions, etc. These methodologies rely on comparing model behavior against the data representations and the physics entities.

- *Model Cross-Validation* could estimate modeling performance and reduce the risk of model over-fitting and under-fitting issues. Commonly used cross-validation techniques include K-fold cross-validation, repeated random subsampling validation, leave-p-out cross-validation, etc.
- *Sensitivity Analysis* seeks to determine the influence of the model inputs and the interactions among inputs on model outputs. It provides a set of practical tools for the robustness of the trained DT models and under what scenarios the model should be trusted. Classical sensitivity analysis includes evaluating the main effects and joint effects and determining the mean response of the model to each input. Besides, the model behavior under extreme conditions should be tested. The model uncertainty should be quantified by varying the model inputs and model parameters over a stated range.

D) DT Online Updating: Analytics for DT online updating can be categorized by spontaneity. Most offline models could be updated manually via just-in-time modeling when it fails the online model evaluation or verification. For specific occasions, there are also automatic online updating methods:

- *Bayesian Learning* (e.g., Bayesian linear regression, Bayesian training of neural networks, etc.) updates the model parameter sequentially as more data is observed. These algorithms are practical when limited data is available at the beginning. The Bayesian model training will gradually improve the model parameter estimation once new data is continuously fed to the algorithm for model updating.
- *Active Learning* refers to a class of machine learning algorithms that retains the model performance by choosing the data that is going to be annotated by human experts. In practice, uncertainty sampling and diversity sampling are two commonly used active learning algorithms. The former identifies the important samples the model is uncertain about. The latter finds the unseen data in the trained model. The identified samples are then sent to human experts for labeling. The model is re-trained or updated after the labels are obtained.

4. Case Studies

Table 4 summarizes the application examples of DT-based intelligent maintenance in manufacturing recently. This section will demonstrate the use of deep learning in intelligence maintenance by using the bandsaw machine case study in Table 4. Also, a cyber-manufacturing system for the intelligent maintenance of distributed machine tool fleets is used to demonstrate the implementation of 5C architecture.

Table 4. List of application examples of DT-based intelligent maintenance in manufacturing

Intelligent Maintenance Use Cases in Manufacturing	Research Description	References
Spindle health (for machine tool)	Machine tool spindle health management under Industrial AI eco-system. Four key enabling technologies are illustrated to develop a systematic health assessment solution for the spindle. The fault diagnostics of ball screws based on monitoring data are developed with an experimental study. A convolution neural network-based method with deep domain adaptation is proposed to validate the transferability of the diagnostic model between sensors.	[2] [4]
Ball Screw	A systematic methodology for ball screw fault diagnosis, health assessment, and RUL prediction is proposed and validated with the two experimental studies.	[5]
Bandsaw machine	A comprehensive methodology for bandsaw machine blade degradation assessment is proposed with a case study. A surrogate model composed of a deep convolution neural network with low-cost process variables is proposed for blade health assessment.	[6]
LCD panel cutting wheel	LCD panel cutting wheel degradation monitoring based on automatic optical inspection. A system methodology is developed through image processing. The feature extraction is based on a specific region of interest (ROI).	[7]
Tool life prediction	The suitability assessment of data from boring tool degradation monitoring is proposed with a case study. The maximum mean discrepancy-based (MMD-based) method is proposed to	[8]

evaluate the detectability, diagnosability, and trendability of data for prognostic tasks.

Automatic bottle capping machine	Bottle capping failure detection and diagnosis based on motor current signal. A fault detection and diagnosis (FDD) tree based on the witness function MMD (WF-MMD) method is used in this application.	[9]
Robotic Arm	RUL prediction of robotic arms under dynamic working regimes. A domain-generalization-adversarial long-short-term memory model is used for RUL prediction.	[42]
Semiconductor Manufacturing	Deposit accumulation assessment based on semiconductor etching process data. A virtual metrology model based on Principal Component Analysis (PCA) with piecewise linear approximation is constructed.	[1]
PHM for Distributed Machine Tool Fleets	A comprehensive PHM framework for large fleets of geographically distributed assets. The proposed framework is composed of three core elements: enhanced fleet-level prognostics, asset risk assessment, and enterprise-level optimization	[11]
Scheduling	A field-synchronized Digital Twin framework for production scheduling was proposed in an experimental study. A Genetic Algorithm-based optimization strategy is used for scheduling optimization.	[10]

4.1. Deep Learning Enhanced Machine Degradation Assessment

Background and Research Objectives: In this case study, a 4-stage roadmap is proposed to demonstrate how deep learning models are applied to establish a physics-information DT band saw degradation assessment and prediction. The developed algorithm is suitable for massive deployment with good prediction accuracy and low deployment costs.

The bandsaw machines are standard tools applied for steel cutting in various manufacturing processes. The wearing condition of the saw blade could impact the cutting performance and cause safety risks, making it one of the most critical components of the bandsaw. It contributes a significant percentage of the bandsaw machine maintenance costs. Common bandsaw blade failure types involve broken teeth, tooth stripping, cracks in the gullets or in the side of the blades, heavy wear on the sides or the tips of the blades, etc. The blade degradation is generally accompanied by excessive vibration and sharp noise from the bandsaw machine. Based on these precursors, experienced engineers could easily tell whether the blade needs to be replaced. In practice, bandsaw blades degrade quickly and must be replaced after a certain amount of operation cycles. Routine maintenance could waste useful life for the less-wearied blades. Leaving blade degradation unattended could also damage the cutting surface, causing scrapped products and even harm the operators and damage surroundings. However, depicting blade degradation conditions and predicting blade remaining life are still blind spots due to high condition monitoring costs. This issue becomes especially prominent when deploying such capability among the massive bandsaw fleet.

To address the complexity of the blade wear and provide intelligent maintenance configurations, degradation assessment analytics, which is an essential block of PHM, is applied to extract meaningful degradation information for the RUL prediction and thereafter to perform the

subsequent maintenance planning. In general, tool wear modeling techniques can be categorized as physics-based or data-driven methods. Physics-based methods estimate or predict the tool life using physical equations, such as Taylor's [43] and Hasting's tool life equation [44]. Physics-based methods have strong generality but cannot incorporate uncertainties in noisy real-world environments. Some crucial model inputs are difficult to measure. Data-driven methods estimate the tool wear from sensor readings using data mining and machine learning techniques, such as Neural Networks, Support Vector Regression, etc. In recent literature, data-driven degradation assessment analytics have been widely studied in various fields, including semiconductor manufacturing equipment [45], machine tools [46], batteries [47], ball-screws [5], rotating machines [48], industrial gas turbines [49], etc. Most of these studies leverage accurate data-driven models, which are, however, based on expensive sensing devices, such as vibration sensors, acoustic sensors, and cutting force sensors. Although these signals are adequate in a lab environment or during prototype development, deployment of these health monitoring technologies requires many expensive sensing devices, data acquisition & storage capabilities, etc. These extra costs prevent these technologies from massive deployment in practice.

Proposed Methodology and Analytics: Fig. 4 demonstrates the proposed 4-stage roadmap for practitioners to develop and deploy the blade degradation assessment models. In stage #1, the purpose is to design the sensing system and data acquisition system to collect sufficient data from a test-bed. This is in preparation for further developing the prototype blades degradation model. A rich amount of sensors are installed so as to satisfy the physics-based modeling needs. The most correlated sensors and features to the blade wear can be identified, and proper testing conditions to study the blade wear can be designed. With the established test-bed, a prototype degradation model is constructed in stage #2 by fusing the physics-based features, experience-based features, and statistical features. The physics-based model is only estimated with data from the typical operating conditions to eliminate the error from the dynamic real operation scenarios. This could also reduce the computational load and enhance the consistency of the degradation depiction. The prototype validation at stage #2 is still based on a limited number of bandsaw machines from the lab environment. At stage #3, the prototype degradation model is further validated on multiple machines from the lab in the field. The data-driven deep learning surrogate model is developed using data only from the cheap sensors and controllers to predict the health indicator (HI) suggested by the output of the physics model in stage #2. The data-driven surrogate model at stage #3 is much more efficient and cheaper than the prototype model at stage #2 since the expensive sensors and data acquisition (DAQ) devices are no longer necessary. At stage #4, the established surrogate model can be easily programmed into the edge devices for continuous blade wear assessment on a massive scale. The prototype model and surrogate model can be further enhanced as data cumulates.

In this case study, the prototype degradation model (or the physics-informed model) was initially developed based on the COSEN C-1000NC automatic horizontal bandsaw machine. The development and validation of the deep learning surrogate model are based on data from the Run-to-Failure (R2F) experiments from three bandsaw machines of the same type. The R2F experiments are conducted on three blades for each bandsaw machine (9 blades in total). Two bandsaw machines are set only to cut round section workpieces to investigate the feasibility of the proposed methodology, and one machine is set to cut two different workpieces to mimic the dynamic scenario. 22 sensor channels are recorded in which the vibration and acoustics sensors are much expensive than all the others.

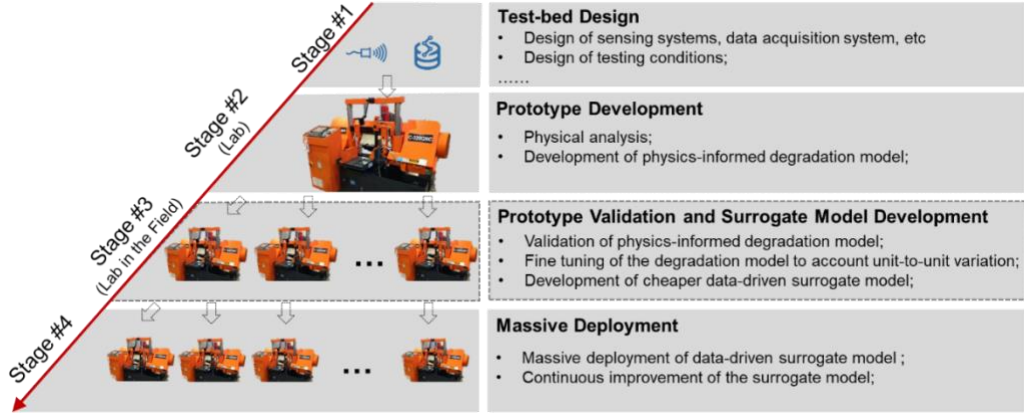


Fig. 4. The 4-stage roadmap for developing and deploying scalable degradation assessment models. Credit to [6].

Table 5. List of sensor channels for prototype modeling [6].

Sensor	Sensor	Sensor	DAQ	Data Velocity	Suitable for Massive
Acoustics	Add-on	Low	High	High (2.56kHz)	No
Vibration	Add-on	High	High	High (2.56kHz)	No
Coolant flow rate	Add-on	Low	Low	Low (2 min)	Yes
Coolant temp	Add-on	Low	Low	Low (2 min)	Yes
Coolant PH value	Add-on	Low	Low	Low (2 min)	Yes
Hydraulic oil temp	Add-on	Low	Low	Low (2 min)	Yes
Ambient temp	Add-on	Low	Low	Low (2 min)	Yes
Blade deviation	Add-on	Low	Low	Low (2 min)	Yes
Blade speed	PLC	No cost	No cost	Low (2 min)	Yes
Downward speed	PLC	No cost	No cost	Low (2 min)	Yes
Cut number	PLC	No cost	No cost	Low (2 min)	Yes
Accumulated cut area	PLC	No cost	No cost	Low (2 min)	Yes
Motor current	PLC	No cost	No cost	Low (2 min)	Yes
Machine ID	PLC	No cost	No cost	Low (2 min)	Yes
Blade ID	PLC	No cost	No cost	Low (2 min)	Yes
Blade TPI	PLC	No cost	No cost	Low (2 min)	Yes
Blade type	PLC	No cost	No cost	Low (2 min)	Yes
Blade brand	PLC	No cost	No cost	Low (2 min)	Yes
Blade material	PLC	No cost	No cost	Low (2 min)	Yes
Cutting material	PLC	No cost	No cost	Low (2 min)	Yes
Work piece height	PLC	No cost	No cost	Low (2 min)	Yes
Work piece width	PLC	No cost	No cost	Low (2 min)	Yes

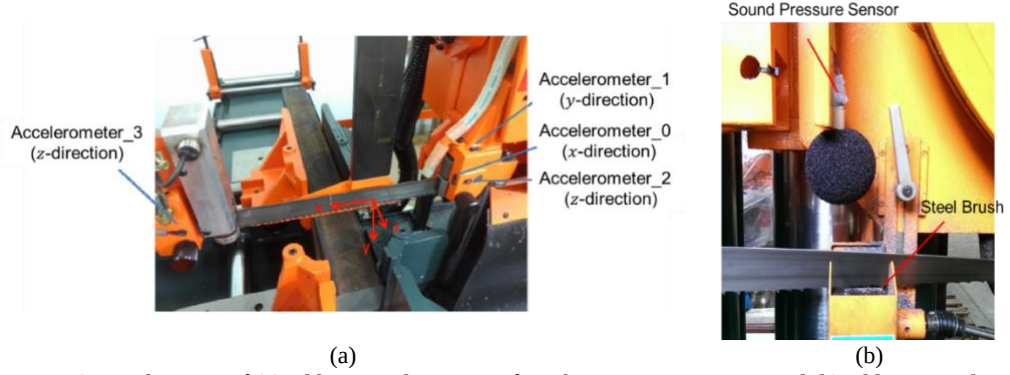


Fig. 5. Sensor location of (a) add-on accelerometers for vibration measurement and (b) add-on sound pressure sensor for acoustic measurement.
Credit to [6].

Table 5 summarizes all the sensor channels from the lab test-bed designed for data acquisition at stage #1. The data collected from the acoustics sensor, vibration sensor, and other sensors are synchronized at first. To collect the physics-based and experience-based features for the prototype model, the add-on accelerometers and add-on sound pressure sensors are set up as seen in Fig. 5 (a) and (b) for vibration and acoustic measurement. Vibration waveforms from these accelerometers are used to understand the vibration modes of the blade while cutting and to locate the characteristic frequency that relates to the blade wear. The physics-based features from the vibration signal are proved crucial in designing reliable features to describe the blade wear. The sound pressure sensor is placed near the steel brush. Inspired by the experience that a constant sharp noise can be heard during the cutting process when the blade is degraded, the noise level at the corresponding frequency band is collected as experience-based features. These two sensors are very informative but expensive compared with the other sensors from the test-bed, including the flow rate, PH level, coolant temperature, motor current, cutting parameters, etc.

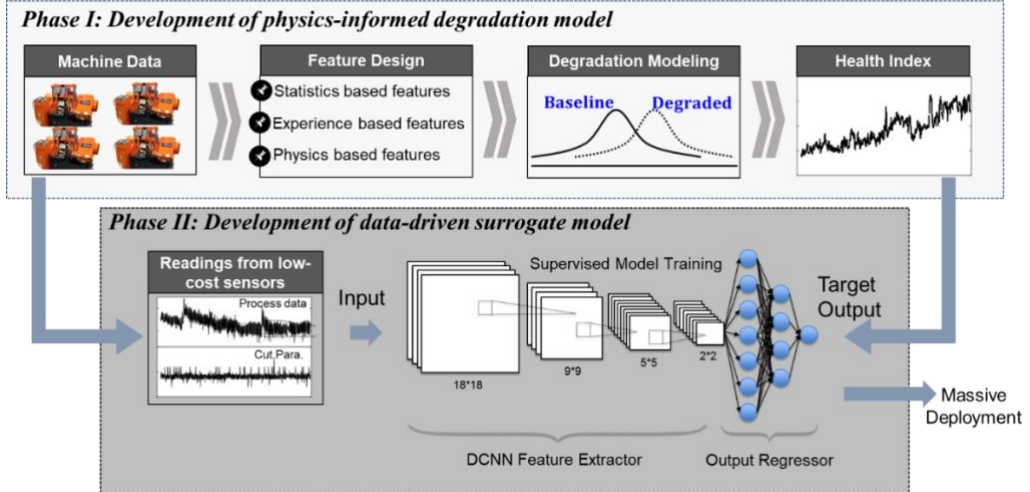


Fig. 6. Methodology for the dual-phase degradation model development.
Credit to [6].

To reduce the cost of the sensing devices and the data storage for massive deployment, we propose to use a deep learning algorithm to extract features from these less informative process variables and develop a cheaper data-driven surrogate model for blade degradation assessment. The methodology for the degradation model established in Fig. 6 has two phases:

Phase I aims to establish a physics-informed degradation model for the blades based on features obtained from vibration and acoustic sensors. The workflow includes data acquisition, feature design, feature selection, and degradation modeling. Feature design relies on physical findings, data expert experience, and analysis insights. Feature selection is based on the feature monotonicity over cutting cycles, the level of noise, and consistency. Once the features are properly extracted from the raw data, the health index that indicates the degradation of the bandsaw blade is further generated from a deviation-based methodology. Data from the first 10% of R2F cycle is considered healthy and the features extracted from it are used to construct the baseline distribution. When the new data come in, the same features are extracted and compared to the baseline feature matrix. The more the current feature matrix deviates from the baseline feature matrix, the more seriously the current blade degraded. Numerous statistical metrics are available to measure such deviation. In this case, Principal Component Analysis (PCA)- T^2 is adopted. The output of this prototype model is the estimated HI appearing as a degradation trace over time.

Phase II aims to train and validate a deep learning-based surrogate model to replace the prototype model developed and validated in Phase I. The surrogate model aims to map the low-cost process variables to the HIs using a deep convolutional neural network (DCNN), which generally contains a multi-layer feature extractor and a regressor at the output layer. Since extracting features from the low-cost process variables relies heavily on domain expertise and traditional features may be contaminated by measurement noise in off-lab scenarios, DCNN provides a powerful solution to address these challenges. By establishing this data-driven surrogate model that is suitable for massive replication, the requirement of expensive sensors and its data acquisition and storage devices could be eliminated thoroughly. Apart from the acoustic and vibration sensors from Table 5, the raw data from other 20 low-cost sensors are used as the input for the DCNN model. The temporal data measurements are cleaned and normalized before being fed to train the model. A sliding window strategy is adopted to format the training sample blocks. As for DCNN model training and validation, data from two blades of each machine (6 blades in total) are used for training and the rest one blade of each machine (3 blades in total named Blade I, Blade II, and Blade III as shown in Fig. 7) are used for validation. In order to demonstrate the superiority of the proposed DCNN for this task, the regression performance is further compared with other popular regression methods: Linear Regression, One-layer neural network (NN), Multi-layer NN, Radius Base Function Support Vector Regression (RBF SVR), Long Short Term Memory networks (LSTM) in terms of root mean squared error (RMSE) and mean absolute error (MAE).

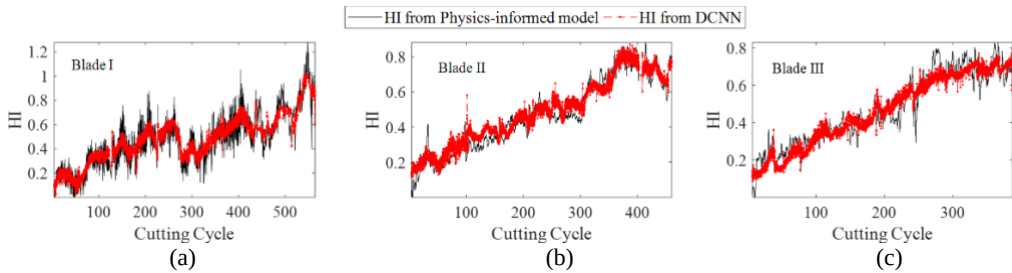


Fig. 7. Blade HI from the physics-informed model and DCNN model for (a) Blade I; (b) Blade II; (c) Blade III.

Credit to [6].

Table 6. Regression performance comparisons of different methods. (std: Standard Deviation) [6].

		Linear Regression		One-layer NN		Multi-layer NN		RBF SVR		LSTM		DCNN	
		mean	std	mean	std	mean	std	mean	std	mean	std	Mean	std
Blade	RMSE	0.276	-	0.358	0.197	0.224	0.084	0.151	-	0.121	0.004	0.084	0.002

I	MAE	0.207	-	0.295	0.194	0.184	0.057	0.121	-	0.095	0.002	0.067	0.001
Blade	RMSE	0.428	-	0.376	0.090	0.293	0.064	0.115	-	0.113	0.006	0.060	0.002
II	MAE	0.406	-	0.350	0.091	0.247	0.061	0.093	-	0.089	0.007	0.050	0.003
Blade	RMSE	0.179	-	0.152	0.025	0.205	0.046	0.083	-	0.080	0.004	0.064	0.002
III	MAE	0.145	-	0.123	0.021	0.167	0.035	0.066	-	0.061	0.004	0.051	0.002

The surrogate regression results on three testing blades are shown in Fig. 7. The estimated blade HI (shown in red) by the proposed DCNN model is very close to the HI given by the physics-informed prototype model (shown in black). The presented high consistency justifies the effectiveness of the proposed methodology of replacing the high-cost sensor-based solution for degradation assessment. The significant fluctuation in Fig. 7. (a) is caused by variations in the size of the workpieces. The variations in the workpiece size may introduce uncertainties to the estimation. But the overall trend of HI can still demonstrate the blade degradation successfully.

The comparative study result of surrogate modeling is summarized in Table 6 where the regression models are fitted ten times for each blade with the random initialization, and the mean values and standard deviations of RMSE and MAE are recorded. The poor performance of the linear regression indicates the relationships between the low-cost variables and HI are non-linear. The poor performance of the one-layer NN further indicates that the target relationship is complicated and cannot be mapped via a shallow network. It is noted that the proposed DCNN method achieves the best performance among all other methods.

In summary, this case study proposes a systematic methodology and roadmap to develop a degradation assessment model for bandsaw machine blades. Without loss of generality, the proposed methodology is applicable in other situations where a low-cost tool wear assessment model is desired for massive deployment. The proposed AI analytics could address the engineering challenges in the proposed intelligent maintenance tasks, making a powerful diagram for massive asset prognostics and health management.

4.2. Cybermanufacturing for distributed assets

This case study demonstrates a framework to build CPS-enabled manufacturing applications for networked fleets of manufacturing assets. Compared to existing Internet-enabled business sectors (e.g., retailing, music, banking, etc.), manufacturing assets are less connected, data values are much less mined, and information is difficult to access. Besides, the production process in distributed factories without considering the machine condition, expands a gap between customer need and asset capability, as shown in Fig. 8.

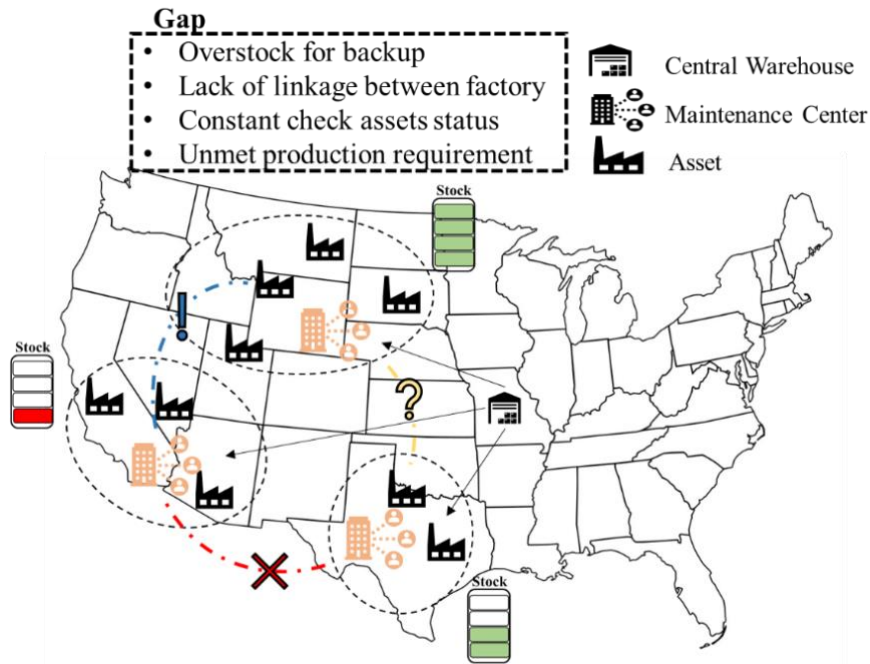


Fig. 8. The current gap between production requirements and remotely distributed machine health. Credit to this chapter.

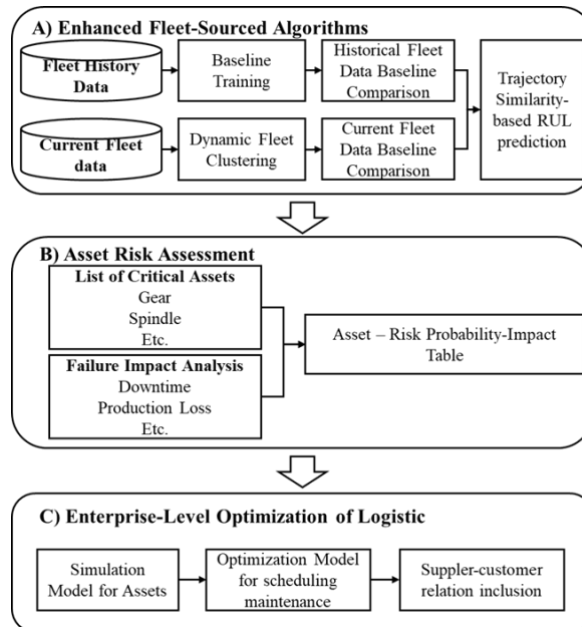


Fig. 9. The Flowchart of Cybermanufacturing Application. Credit to this chapter.

Despite these challenges, manufacturing data are more structured and the operating environments are comparatively controlled. In this section, the proposed methodology takes advantage of the inherited similarity between these assets to perform advanced data-driven prognostics for asset health condition prediction and discover higher-level issues for similar assets (fleet) to support

management decision-making. Users could access actionable predictive information on asset conditions through mobile applications. The whole manufacturing enterprise is more responsive and resilient to disturbances, and production assignments can be more accurately distributed to meet production requirements better. Therefore, the flowchart of this application can be illustrated in Fig. 9.

Objective and Research Motivation: The motivation for cyber manufacturing comes from the following three perspectives.

- a) With mobile applications proven to be revolutionary for various service industries, increased information transparency and connectivity in manufacturing have been called for to foster better responsiveness in logistics and resource allocation, as well as increased resilience in production.
- b) Collecting enough data is becoming less of an issue for manufacturing than dealing with data. Some enterprises are now generating millions of data every day. The development of advanced data-driven technologies that can adapt to this Big Data environment has become necessary to convert data into meaningful and actionable information.
- c) Current business decisions are usually top-down and rarely consider the health condition of assets and systems. A bottom-up information flow is needed to inform and improve decision-making so that manufacturing production assignments can be optimized according to the asset health conditions.

Based on the abovementioned motivation, the objective is to develop mobile applications that can reflect single asset health conditions based on fleet-sourced prognostic information. In this light, the objectives are summarized as follows:

- a) To leverage the similarity and peer-to-peer based methods to build a dynamic fleet-based prognostic model;
- b) To devise a risk assessment methodology for prognostics and performance evaluation;
- c) To develop fleet-sourced, enterprise-wide mobile applications for real-time access to predictive manufacturing information.

Proposed Framework:

A) Enhanced Fleet-Sourced Algorithms: Clustering can be applied to group assets under similar conditions and generate local baselines based on similarities between machine conditions. The similarity of machines is evaluated from the perspectives of machine make and model, working conditions, and operation configurations. Two major causes of machine behavior changes are varying working regimes and the degradation/fault condition of machines. Working regime changes mainly result from: (a) Environmental factors like seasons or weather and (b) customized operation condition for specific machine tasks. The core way to differentiate a cluster pattern change is to judge the conformity of the current clustering structure. Therefore, the general term ‘similarity characterization’ is used to define the algorithms that characterize the adjacency of members within one cluster. In the training phase, the in-cluster similarity of each cluster is computed in advance. Then, owing to the computed in-cluster similarity, a threshold will be set up through a user-defined confidence level from the statistical distribution or simple statistics. The current in-cluster similarity will be re-evaluated by the threshold before judging the current asset health condition. If the current in-cluster similarity is over the threshold, the original clustering structure has to be re-defined. All in all, the dynamic clustering approach will enable the fleet-level PHM to quickly adjust to the complicated working condition changes and robustly offer consistently accurate PHM results. Finally, a confidence value (CV) can be calculated, and the related equation can be found in the literature [50]. When the CV of an asset is calculated, future performance will be evaluated to accomplish the risk assessment and enhance operation management. The similarity-based method can be used to predict the 1-D CV [51]. After the baseline training phase, the degradation condition of performance will be extracted from the history data for each asset and accumulate a library of

degraded trajectories. Several algorithms are suitable for employment, such as kernel regression or the hidden Markov model. When the CV of the current assets over time is computed, the prediction can be obtained by comparing the similarity between the current trend of CV and curves in the library. Finally, RUL estimation can be generated by calculating a similarity-weight sum of all the trajectories. The related similarity metric can be found in the literature[51].

B) Asset Risk Assessment: From the perspective of statistics, CV can be regarded as the probability of machine failure. To thoroughly evaluate the risk of a potential issue, making a list of critical assets and the corresponding impact is important. As shown in Table 7, the “Asset-Risk Probability-Impact” table can be seen as an effective tool to quantify the risk of critical assets.

Table 7. Risk analysis tool: Asset – Risk Probability – Impact Table

		Probability of Failure	Downtime (Days)	Production Loss/Min	...
Assets	Gear	0.8	10	10	...
	Spindle	0.65	20	11	...
	...	...	...	...	...

In the case of an asset containing several different failure modes, each failure mode should be put as a single item in the “Asset” column instead of the “Failure Mode” column because multiple failure modes sometimes happen at the same time. After obtaining the table, the quantitative risk assessment will be calculated by following the equation:

$$RI_j(t) = CV_j(t) \times DT_j \times PL_j \times IF \quad (1)$$

where RI_j is the risk index (RI) of asset j , $CV_j(t)$ is the confidence value of asset j over time t , and DT_j and PL_j are downtime and unit time production loss for asset j . Impact factors (IF) can be safety, the severity of the hazard, and so on. This will serve as the foundation for operation management, logistic prioritization, maintenance scheduling, and production assignment decision support.

C) Enterprise-Level Optimization of Logistics: The enterprise-level optimization of logistics can be formalized as a stochastic problem with a user-defined cost function that simultaneously considers the availability, asset function, and the impact of downtime. An illustration of geographically distributed assets, served by a set of maintenance centers where maintenance personnel resides. Overall, the proposed method in this section will be executed through the following steps:

- Build up the simulation model of a target system for optimization in the next step. The simulation model includes different models of assets as sets of multiple components [52], component reliability, and supplier-customer relation across the enterprise.
- Develop an optimization model for contemporary optimization of inventory levels and maintenance schedules in the enterprise. Decision-variables will contain inventory levels of spare parts and raw materials, as well as maintenance schedules of component or factory, which is optimized through simultaneous considerations of logistic and production operational cost.
- Involve enterprise-wide, dynamic optimization of supplier-customer relations and factory-specific production to respond to changes in the condition and performance of various parts in the enterprise. This step will allow the enterprise to reconfigure itself by adjusting underperforming nodes while additionally loading nodes with better capability.

Demonstration of Machine Tool Fleets: The proposed framework of fleet-sourced cyber manufacturing applications can be summarized in the diagram shown in Fig. 10. After identifying

the critical parameters to collect from a machine, features that contain diagnostic information will be extracted based on experience and domain knowledge. Dynamic fleet clustering will then be performed based on asset similarity, where the similarity of machines will be evaluated according to the machine make and working environments, and operating configurations. Anomaly detection and predictive modeling in local clusters will be more accurate than global independent modeling[50], and information from each asset will serve as a cyber twin model of the physical machine. The cyber twin model will be presented in the mobile applications to be developed so that operators or managers can keep up with the changes in asset health conditions from anywhere at any time. The enhanced fleet-based prognostics and asset risk assessment will be elaborated in the next few subsections, followed by the description of mobile app development.

A) Mobile Application Development: we intend to develop mobile applications for cyber-manufacturing to provide users with the ability to stay in touch with predictive asset health information. Mobile application development is divided into two major tasks: Back-end and Front-end development. These two sections are highly tied together as the front-end features mandate certain capabilities on the back-end. A detailed description of these tasks is presented in the rest of the section. After initial development, the beta version of the application will be shared with a limited number of member companies through a test flight program to receive their feedback continuously. During the test and improvement phase, we can make sure that the application fulfills enterprise requirements.

B) Back-end development: The back-end of any web or mobile application is the platform on which the application interacts. Depending on the amount of processing needed, the back-end side can be a single web server with a simple web service or a gigantic cloud server with complex data processing methods and web services. This task can be divided into three major subtasks as follow:

C) Platform Development: In real-world applications, cyber-manufacturing platforms will continuously receive massive amounts of data from fleet members. The amount of incoming data is proportional to the number of active fleet members, and instantaneous data processing is required to keep the whole network of cyber-machines updated. These requirements can be fulfilled by using a proper distributed system as the foundation of the platform. Distributed platforms, such as Hadoop, have the flexibility to increase processing power and data storage capacity.

D) Analytical Cloud Software Development: Once the infrastructure is in place, data processing algorithms have to be modified to be compatible with running over distributed systems. These algorithms will be part of a larger cloud-based software program that maintains the flow of information, data processing jobs, user management, etc. This software program is the core of the mobile application.

E) Web Service, Data Transfer, and Security: To maintain the security of the data being transmitted between applications and servers, the proper protocols and web services are usually required. Communications have to be encrypted over HTTPS sessions. At the same time, modern web services such as REST (Representational State Transfer) or SOAP (Simple Object Access Protocol) are used to maintain the data transfer.

F) Front-end development: Front-end development consists of developing the application flow, designing the user experience, and interacting with the back-end. One of the important and challenging tasks in designing the interface is to consider presenting a vast amount of information generated by the fleet-based analysis in an optimized manner so it would be easily recognizable by the users. On the other hand, as users with different corporate roles use the application, appropriate sets of information have to be presented to each user. Therefore, custom-designed visualization methods have to be used to convey more information to the user in a more abstract matter. Besides the interface features, the application has to maintain a secure connection to the back-end service to receive information. The connection can be made over a corporate virtual private network (VPN) and through encrypted secure socket layer (SSL) connections.

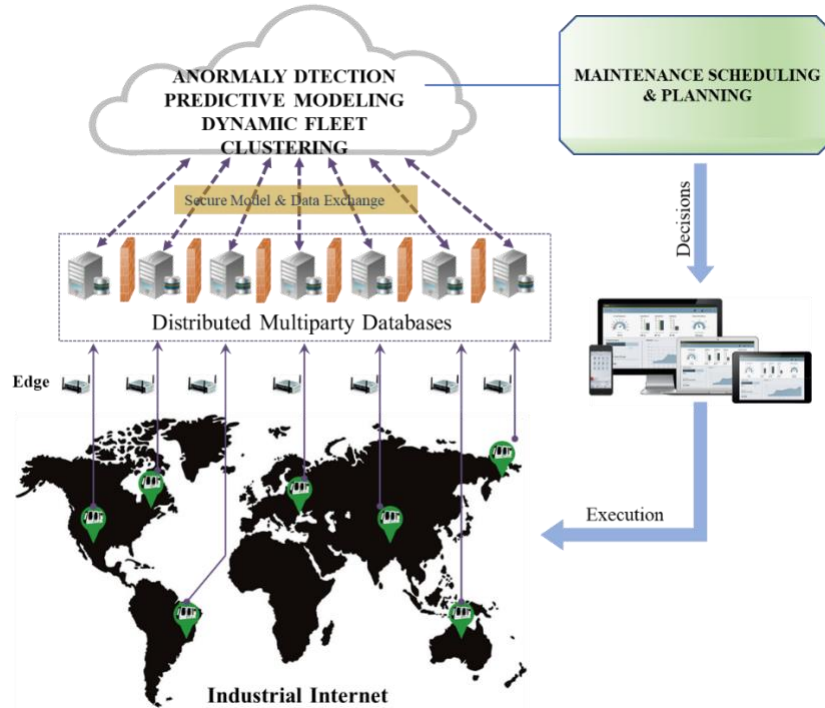


Fig. 10. The demonstration of machine tool fleets.
Credit to this chapter.

5. Conclusion

With the expanding demand for high quality, flexibility, and efficiency in manufacturing, smart factories hold the vision of achieving timely acquisition, distribution, and utilization of information from components, machines, and shop floors. The CPS paradigm and its enabling technologies serve as a one-stop solution throughout the manufacturing and maintenance processes. This book chapter summarized the CPS frameworks, its critical enabling technologies, and its core technology digital twin regarding smart manufacturing and maintenance.

This chapter first illustrates the innovative CPS framework for maintenance and service innovation in manufacturing, see Fig. 1. The proposed CPS framework consists of 5 layers: 1) the connection layer collects the real-time machine data from sensors, machine controllers, and quality inspections for subsequent analysis; 2) The conversion layer prepares the machine data into representative features and meaningful information for knowledge discovery and asset management. 3) The cyber layer performs data modeling, analysis, and prediction in a networked environment that overcomes the physical limit to connect the distributed assets of the enterprise and makes fleet Analysis or big data analysis possible. 4) The cognition layer provides decision supports and synchronizes the decisions and information to relevant parties through well-design visualization tools or infographics. 5) The configuration layers act as a control system that executes corrective or preventative decisions.

Key enabling technology of the proposed framework is then described, which includes data technology, analytics technology, operation technology, and platform technology. Data technology refers to the technologies that enable the acquisition of useful data from machines, which include advanced sensing and monitoring, data indexing, transmission, and storage, data quality calibration, big data representation and visualization, data security, privacy protection, etc. Analytics

technology converts the data from different machines and different sources into meaningful (or predictive) information that can be understood and utilized by human experts. Data analytics include signal processing, statistical analysis, detection, classification, prediction, clustering, relationship-mining, causal analysis, inference, modeling of complex and dynamic systems, etc. Operation technology refers to the decision support system that recommends corrective and preventative actions based on meaningful information from machine data. The operation technology includes efficient large-scale planning and scheduling program, manufacturing system modeling and simulation tools, real-time optimization and decision-making analytics, etc. Platform technology refers to hardware and software infrastructures concerning computation, simulation, visualization, data storage, and servitization, including edge and cloud computing, automation and control, machine-to-machine communication, human-machine, etc.

The core element of the proposed CPS framework is a DT which is a dynamic digital replica of the physical manufacturing systems, processes, and services. The DT can be designed for diversified purposes such as predictive maintenance, design improvement, intelligent control, knowledge retention and expanding, work-in-progress scheduling and planning, etc. In this book chapter, a systematic methodology for DT development and deployment is described. The proposed methodology has an offline phase that focuses on model development and an online phase for model deployment and online updating. The data analytics that is needed for DT development, DT adaptation, DT validation and verification, and DT online updating are also reviewed comprehensively.

Two real-world use cases are provided to demonstrate the use of advanced data analytics and CPS technology in maintenance and service innovations. Cutting edge deep learning algorithms are employed in this case to process the real-time machine data (e.g., sound pressure, vibration, controller data, etc.) into the saw blade's remaining life, based on which maintenance activities can be scheduled before failure occurs. The second case demonstrates a cyber-manufacturing system for predictive maintenance of distributed CNC machines. Spindle health monitoring algorithms are deployed to an edge device that connects to the internet. Then the health information from the distributed machine fleets is aggregated in the cloud for maintenance planning and scheduling.

While research on advanced data analytics and CPS have been fruitful, challenges still exist in the widespread adoption of these technologies in the manufacturing industry. It has been found that most manufacturers, particularly Small and Medium Enterprises (SMEs), are not fully aware of the technical risks and potential benefits, and they are unwilling to invest in expensive technical infrastructures and required R&D without a proven return on investment [53]. Cybersecurity and privacy concerns further diminish the incentives to adopt these technologies. Furthermore, another major barrier to the implementation of new technologies (or solutions) is often logistic factors (e.g., knowledge, workers, training, security, etc.) rather than the technology itself. To overcome these barriers, Workforce empowerment (e.g., training, culture shift, ergonomics to match the solution to worker capabilities, knowledge retention, expansion, etc.) is pivotal.

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